

IEDA 4000H

Homework 4

Due Date: Nov 16 (Sunday)

1. Implement a Python function to price a European call option using the multi-period binomial lattice model described in Lecture 14. The model assumes the underlying asset price evolves over N periods, starting from initial price S_0 , with up-factor $u > 1$, down-factor $d < 1$ (where $d = 1/u$ for simplicity, as suggested in the lecture for small time steps), and risk-free rate r per period (so $R = 1 + r$). The risk-neutral probabilities are $p_u = \frac{R-d}{u-d}$ and $p_d = 1 - p_u$. Use backward dynamic programming to compute the option value at each node, as in the recursion:

$$v(k, j) = \frac{1}{R}(p_u v(k+1, j+1) + p_d v(k+1, j)),$$

where k is the time step (from 0 to N), j is the number of up-moves (0 to k), and at maturity ($k = N$):

$$v(N, j) = \max(u^j d^{N-j} S_0 - K, 0)$$

with strike price K .

Input Parameters (all provided in the function call):

- S_0 : Initial asset price (float).
- K : Strike price (float).
- N : Number of periods (int).
- r : Risk-free rate per period (float).
- u : Up-factor (float, $u > 1$). If not provided, compute as $u = e^{\sigma\sqrt{\Delta}}$ where $\Delta = 1/N$ and σ is volatility (provide σ if using this; default to given u).
- d : Down-factor (float, $d < 1$). Defaults to $1/u$.

Requirements:

- Use NumPy for array operations (import numpy as np).
- Implement without recursion (use iterative backward induction with a 2D array for the lattice).
- Output the option price $v(0, 0)$.
- Handle cases where p_u or p_d is not in $[0, 1]$ by raising a `ValueError` (indicating arbitrage, per Lecture 13).
- Test with: $S_0 = 100$, $K = 105$, $N = 3$, $r = 0.05$, $u = 1.1$.

Additional Information: Volatility σ for parameter estimation is the annual standard deviation of log-returns. Time to maturity is 1 year, divided into N periods.

2. Consider a simple single-period market from Lecture 13. The underlying stock XYZ has current price $S_0 = 50$. In one year, with two equally likely scenarios (probability 0.5 each, though not used for pricing):

- Up scenario (Ω_u): Stock price $S_1^u = 70$.
- Down scenario (Ω_d): Stock price $S_1^d = 40$.

The risk-free rate over the year is $r = 0.10$, so $R = 1 + r = 1.10$ (lending/borrowing at this rate). We want to price a European call option on XYZ with strike price $K = 55$, which pays $\max(S_1 - 55, 0)$ at maturity.

- (a) Use the replication principle to find the fair price C_0 of the call option. Form a portfolio of Δ shares of the stock and B dollars in the risk-free asset (cash) such that the portfolio payoffs match the option payoffs in both scenarios:

$$\Delta \cdot 70 + B \cdot 1.10 = 15 \quad (\Omega_u : 70 - 55 = 15)$$

$$\Delta \cdot 40 + B \cdot 1.10 = 0 \quad (\Omega_d : 40 - 55 < 0 \implies 0)$$

Solve for Δ and B , then $C_0 = \Delta \cdot 50 + B$. Also, derive the risk-neutral probabilities $p_u = \frac{R-d}{u-d}$ and $p_d = 1 - p_u$ (where $u = 70/50 = 1.4$, $d = 40/50 = 0.8$), and verify that $C_0 = \frac{1}{R}(p_u \cdot 15 + p_d \cdot 0)$.

- (b) Suppose the call option is observed in the market at a price of $C'_0 = 8$ (higher than your answer from Part (a)). Does this create an arbitrage opportunity? If yes, specify the type (A or B from Lecture 13) and construct an explicit trading strategy: state the positions in stock, cash, and option, the initial cash flow, and the payoffs in both scenarios.

Additional Information: In this complete market (two assets, two scenarios), replication uniquely determines the price. Arbitrage exists if the market price deviates from the replicated value, allowing a riskless profit.